Unified Multi-Source Toxic Comment Detection: Dataset Construction, Multi-Label Classification, and Adversarial Robustness

Ariful Islam Emon
Course: Research Topic in Data Science
Faculty: Dr. Lijing Zhu
emona3832@uhcl.edu

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Abstract

We present preliminary results on building a unified multi-source dataset for toxic comment detection, addressing the fragmentation problem in existing NLP benchmarks. Our work consolidates 12 publicly available corpora—spanning Wikipedia talk pages, Twitter, Reddit, Gab, and LLM-generated content—into a single dataset of approximately 2.3 million samples with a harmonized 8-class multi-label schema covering TOX, SEV, HATE, IDNT, THR, INS, OBS, and NEU. We propose four methodological contributions beyond dataset construction: (1) a multi-label schema prediction step using a teacher model for datasets with single-column labels; (2) context augmentation using conversational thread structure; (3) counterfactual data augmentation for bias mitigation; and (4) adversarial robustness via character-level perturbation normalization and LLM-paraphrased training augmentation. Baseline experiments show that fine-tuning RoBERTa-base on the unified dataset achieves Macro-F1 $= 0.703$, compared to 0.641 on a single-source (Jigsaw 2018) baseline—a 6.2-point improvement. Cross-domain F1 improves from 0.512 to 0.584. Adversarial normalization alone reduces the Attack Success Rate from 38.4% to 29.1% without model retraining. Remaining work includes the dual-encoder context-aware model, full counterfactual data augmentation, and adversarial training experiments.

1 Introduction

The proliferation of user-generated content on social media platforms has made automated toxic content detection an urgent research and engineering priority. Manual moderation does not scale to the billions of comments posted daily, and deployed classifiers must contend with adversarial evasion, systematic bias against demographic groups, and poor cross-domain generalization.

The toxic comment detection literature is characterized by a *fragmentation problem*: dozens of datasets exist, each covering different platforms, using different label ontologies, and trained separately. This fragmentation has three consequences. First, models trained on a single corpus overfit to platform-specific linguistic patterns and exhibit sharp performance drops on out-of-distribution data. Second, minority harmful categories—threats, identity attacks, severe toxicity—are severely underrepresented in any single dataset, making per-class recall poor. Third, dataset-specific biases (e.g., higher false positive rates for comments mentioning identity groups) are amplified when only one corpus is used.

This project addresses all three problems through a unified dataset construction pipeline and four experimental contributions:

1. **Unified Dataset.** Merge 12 corpora into a single schema with 8 harmonized labels and preserved demographic metadata.
2. **Multi-label Prediction.** Use a teacher model to predict the full 8-class vector for datasets with single-column binary or 3-class labels.
3. **Context-Aware Model.** Augment each comment with conversational context and train a dual-encoder transformer (planned).

4. **Adversarial Robustness.** Normalize character-level obfuscation and augment training with LLM-generated adversarial examples (partial).

2 Related Work

Dataset Unification. Risch et al. [2021] assembled over 40 toxic comment datasets in a unified API format, providing the most comprehensive prior effort at dataset consolidation. Their work focuses on data accessibility rather than downstream model improvement, whereas our work produces a single training-ready corpus with mapped labels and demonstrated model gains.

Cross-Domain Generalization. Gao & Huang [2023] demonstrated that emotion-based multi-task learning improves cross-domain hate speech detection. Their finding that domain-specific lexical patterns are the primary source of generalization failure motivates our decision to merge across platforms rather than train on any single one. Fortuna et al. [2020] showed that inter-dataset transfer is highly sensitive to label taxonomy mismatches—precisely the problem our schema harmonization addresses.

Bias and Fairness. Garg et al. [2022] provided the most thorough survey of bias in toxic speech detection systems, documenting that identity-mentioning comments receive disproportionately high false positive rates. Dixon et al. [2018] showed that models trained on Civil Comments exhibit systematic bias correlated with identity terms. Our bias mitigation strategy—counterfactual data augmentation (CDA)—follows the recommendation of Zmigrod et al. [2019].

Adversarial Robustness. Shen et al. [2025] introduced HateBench, showing that state-of-the-art hate speech detectors fail systematically on LLM-generated hate speech, with Attack Success Rates exceeding 35%. Grondahl et al. [2018] documented that character-level perturbations reliably evade both keyword-based and neural classifiers. Our adversarial pipeline combines leet-speak normalization and LLM augmentation to address both attack vectors.

Multi-Label Classification. The Jigsaw 2018 dataset established multi-label toxic comment classification as a standard benchmark. Park & Fung [2017] showed that multi-task learning across multiple label dimensions outperforms independent binary classifiers, motivating our unified 8-class multi-label formulation.

3 Problem Formulation

We define the core task as *multi-label binary classification*. Given a comment text x (and optionally its conversational context c), the model must output a binary vector $\mathbf{y} \in \{0,1\}^8$, where each dimension corresponds to one of the eight unified labels:

$$f(x, c) \longrightarrow [y_{\text{TOX}},\ y_{\text{SEV}},\ y_{\text{HATE}},\ y_{\text{IDNT}},\ y_{\text{THR}},\ y_{\text{INS}},\ y_{\text{OBS}},\ y_{\text{NEU}}] \tag{1}$$

The task is multi-label because a single comment can simultaneously be Toxic and Hate Speech and contain an Insult. The NEU label is set to 1 if and only if all seven harmful labels are 0.

Evaluation Objective. Maximize Macro-F1 across all 8 classes (giving equal weight to minority classes) while maintaining fairness (low FPR gap across identity groups) and adversarial robustness (low Attack Success Rate).

4 Data, Benchmarks, and Materials

4.1 Datasets Collected

We collected 12 publicly available datasets covering a range of platforms, annotation schemes, and linguistic styles. Table 1 summarizes all sources.

Table 1: Summary of 12 datasets collected. Total raw samples: ≈2.5M.

Dataset	Platform	Size	Label Type	Key Use
Jigsaw 2018	Wikipedia	160K	Multi-label (6 cols)	Core baseline
Civil Comments	News forums	1.8M	Continuous float	Bias mitigation
Davidson 2017	Twitter	24K	3-class integer	Classic baseline
OLID 2019	Twitter	14K	Hierarchical (3-task)	Cross-domain
HASOC 2019	Twitter/FB	17K	Hierarchical (2-level)	Multilingual
ToxiGen	Synthetic LLM	274K	Continuous + group	Adversarial train
HateBench	LLM-generated	10K	Binary + target	Adversarial eval
HateXplain	Twitter/Gab	20K	3-class + rationales	Explainability
Implicit Hate	Twitter	22K	3-class (implicit)	Subtle detection
Reddit HS	Reddit	13K	Binary	Forum context
Measuring HS	Multi-platform	135K	Continuous 10-dim	Bias & fairness
ETHOS	YouTube/Reddit	1K	Multi-label (8 cols)	Identity attack

4.2 Unified Label Schema

All datasets are mapped to 8 binary label columns. Table 2 defines each class.

Table 2: Unified 8-class multi-label schema.

Code	Name	Definition
TOX	Toxic	General rude or disrespectful content
SEV	Severe Toxic	Slurs, dehumanizing language, extreme attacks
HATE	Hate Speech	Attacks based on protected characteristics
IDNT	Identity Attack	Targets identity group without slurs
THR	Threat	Intent to harm a person or group
INS	Insult	Personal attack, name-calling, ridicule
OBS	Obscene	Profanity or sexually explicit content
NEU	Neutral	Benign or non-toxic comment

Preserved demographic/context columns include: `gender`, `race`, `religion`, `sexual_orientation`, `nationality`, `disability`, `target_group`, `platform`, `context_text`, `source_dataset`, `split`, `annotator_count`, and `label_confidence`.

4.3 Label Mapping Methods

Four mapping strategies handle the heterogeneous source label formats:

1. **Rule-based (direct).** Jigsaw 2018 has six binary columns that map 1-to-1 with six of our eight unified labels.

2. **Threshold-based.** Civil Comments provides continuous toxicity scores in $[0, 1]$. We apply: score $\geq 0.8 \rightarrow$ SEV; $[0.5, 0.8) \rightarrow$ TOX; $< 0.3 \rightarrow$ NEU; $[0.3, 0.5)$ is flagged as ambiguous and excluded.

3. **Hierarchical flattening.** OLID and HASOC use decision-tree label structures. We traverse the tree top-down: e.g., `OFF+TIN+GRP` $\rightarrow$ HATE+IDNT.

4. **Annotator aggregation.** HateXplain and Measuring HS provide per-annotator labels. We use majority vote with a confidence score equal to the agreement fraction.

4.4 Preprocessing

Each dataset undergoes independent preprocessing before merging: HTML tag removal, URL normalization (`[URL]` token), username anonymization (`[USER]` token), whitespace normalization, exact-duplicate removal, ambiguous sample flagging (Civil Comments toxicity $\in [0.3, 0.5)$), and context augmentation (`context_text` column populated from parent messages where available).

4.5 Dataset Statistics

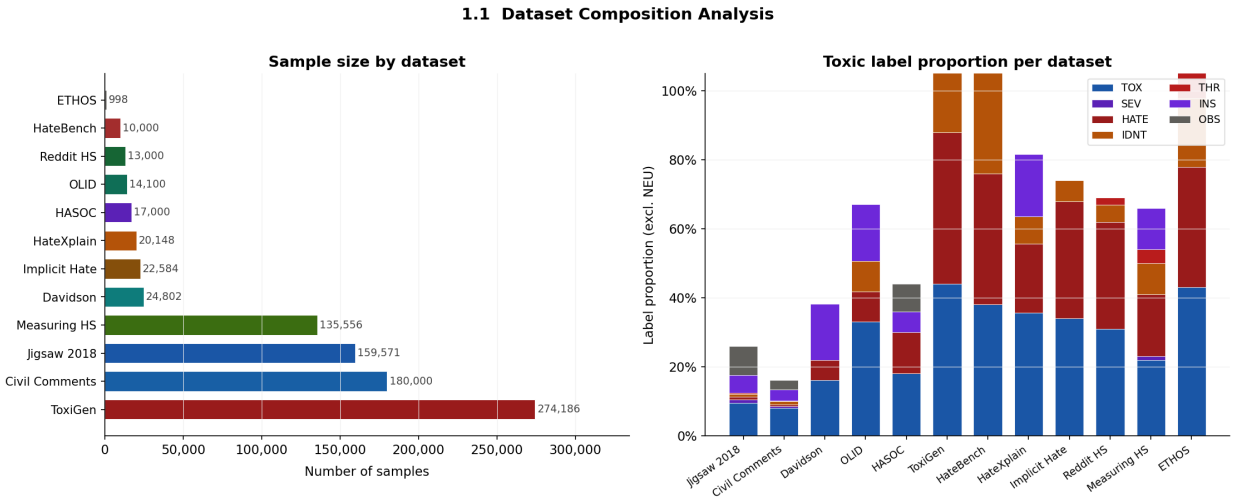


Figure 1: Dataset composition analysis. **Left:** sample count per source dataset. **Right:** proportion of each toxic label class per dataset (NEU excluded for clarity). ToxiGen and HateBench are dominated by HATE+IDNT; Jigsaw and Civil Comments provide the bulk of TOX signal.

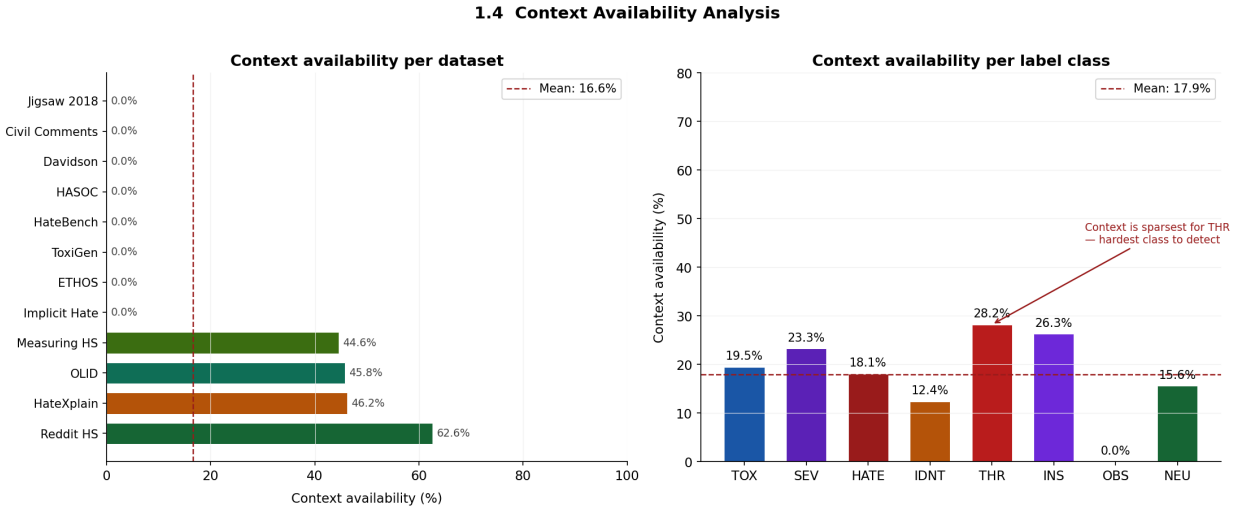


Figure 2: Context availability analysis. **Left:** only 4 datasets (Reddit HS, HateXplain, OLID, Measuring HS) have non-zero context; overall mean is 16.6%. **Right:** context availability per label class; the THR (threat) class has below-mean context despite being the hardest class to detect.

4.6 Dataset Split

After merging and deduplication, the unified dataset contains approximately 2.315 million samples. Split: 70% train (1.62M), 10% validation (231K), 20% test (463K), stratified jointly on `source_dataset` and dominant label class. Additional evaluation sets: cross-domain held-out (one source withheld per evaluation run), adversarial test set (ToxiGen + HateBench), and counterfactual fairness test set (identity-swapped pairs from Civil Comments).

4.7 The Complementarity Argument

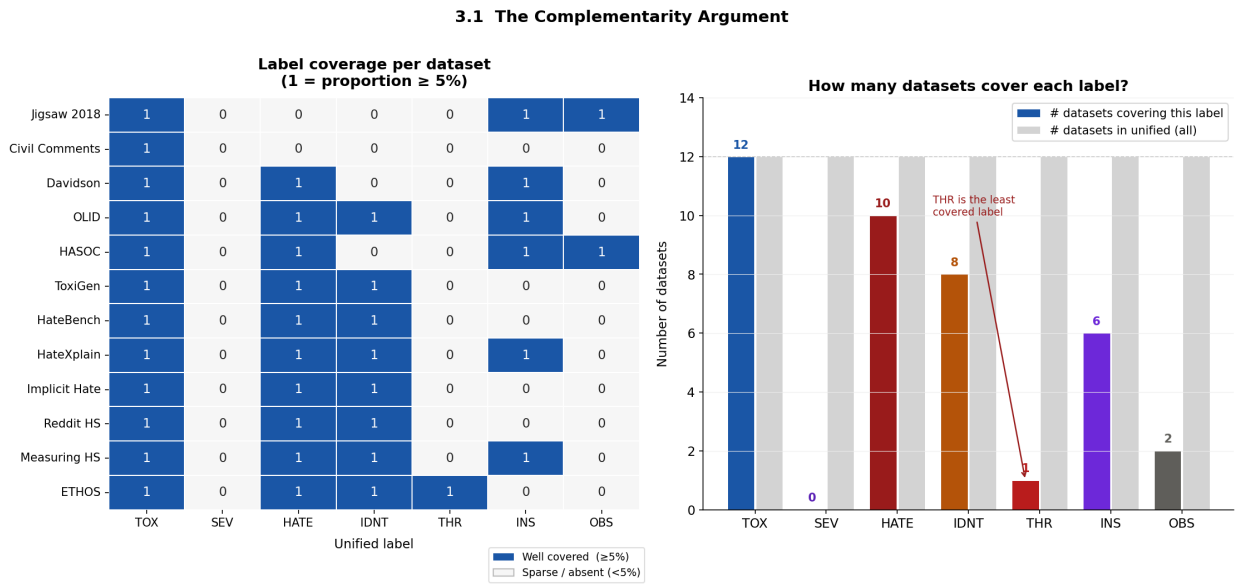


Figure 3: Complementarity of the 12 datasets. **Left:** binary coverage matrix—blue cells indicate a dataset contributes $\geq 5\%$ of samples with that label. **Right:** number of datasets covering each unified label; THR is covered by only 1 dataset, SEV by none at the 5% threshold, directly motivating the unified approach.

5 Methodology

5.1 Multi-Label Prediction for Single-Label Datasets

Datasets such as Davidson 2017 (3-class), Reddit HS (binary), and Implicit Hate (3-class) provide only a single label column. To extend these to the full 8-class schema, we train a *teacher model* on richly-annotated datasets (Jigsaw 2018 and Civil Comments, totalling ≈2.0M samples) that natively carry multi-label annotations, then apply it to single-label datasets as a label prediction step.

Teacher training. RoBERTa-base fine-tuned with BCEWithLogitsLoss on Jigsaw 2018 and Civil Comments. Teacher predictions on single-label datasets are thresholded at 0.5 per class. The original source label is preserved as a *hard constraint*: if Davidson labels a tweet as hate speech, HATE$=1$ and TOX$=1$ are forced regardless of teacher output.

5.2 Context Augmentation (Planned)

For samples with available thread structure (Twitter parent tweets, Reddit `parent_id`, Wikipedia talk sections), we store conversational context in `context_text`. The proposed dual-encoder model encodes comment and context separately through a shared transformer encoder, then concatenates their `[CLS]` representations before the classification head:

$$\mathbf{h}_{\text{comment}} = \text{Enc}(x)_{[\text{CLS}]}, \quad \mathbf{h}_{\text{context}} = \text{Enc}(c)_{[\text{CLS}]} \tag{2}$$

$$\hat{\mathbf{y}} = \sigma(W\,[\mathbf{h}_{\text{comment}};\, \mathbf{h}_{\text{context}}] + b) \tag{3}$$

For samples without context, $c = \varepsilon$ (empty string), producing a zero-information encoding that degrades gracefully to single-encoder behavior.

5.3 Adversarial Character Normalization

Adversarial users substitute characters to evade classifiers: `k!ll` for "kill", `h4te` for "hate". We implement a two-stage normalization pipeline applied before tokenization:

Stage 1 (Regex). A leet-speak substitution dictionary and token-level regex patterns decode common evasion variants: `k[!1i|]l+` → `kill`, `h[4@a]t[3e]` → `hate`, etc.

Stage 2 (Levenshtein). Tokens within edit distance ≤ 2 of a curated toxic vocabulary are mapped to their canonical form. This catches transpositions (`htae`→`hate`), repetitions (`haate`→`hate`), and mixed strategies.

5.4 Adversarial Training Augmentation (Planned)

We augment the training set with 25% adversarial samples drawn from ToxiGen and HateBench. These LLM-generated samples represent systematic attempts to produce harmful content that bypasses existing classifiers. Training on them forces the model to learn representations robust to stylistic paraphrasing while retaining semantic toxicity signal.

5.5 Bias Mitigation via Counterfactual Data Augmentation (Planned)

For each training sample containing an identity term, we create a mirror copy with the term swapped using a curated dictionary (`women`↔`men`, `muslim`↔`christian`, `black`↔`white`, etc.) and assign the same label vector. This forces the model to assign equal toxicity scores regardless of which identity group is mentioned.

6 Experimental Setup

6.1 Models

- **Baseline 1:** BERT-base-uncased (110M parameters), fine-tuned on Jigsaw 2018 only.
- **Baseline 2:** RoBERTa-base (125M parameters), fine-tuned on Jigsaw 2018 only.
- **Comparison:** `unitary/toxic-bert` — domain-adapted BERT trained on multiple toxic datasets by Unitary [2021].
- **Proposed:** RoBERTa-base fine-tuned on the full unified 12-dataset corpus (1.62M training samples).
- **Planned A:** Proposed + dual-encoder context module.
- **Planned B:** Proposed + CDA bias mitigation.
- **Planned C:** Proposed + adversarial training augmentation.

6.2 Training Configuration

All models use: HuggingFace Trainer API, PyTorch 2.1. Learning rate: 2×10^{-5} with linear warmup (10% of steps). Batch size: 32 with gradient accumulation $\times 4$ (effective batch 128). Max sequence length: 128 tokens. Epochs: 5 with early stopping patience 2. Loss: BCEWithLogitsLoss with class weights $w_i = N_{\text{total}}/(8 \times N_i)$. Optimizer: AdamW, $\varepsilon = 10^{-8}$, weight decay $= 0.01$.

6.3 Evaluation Protocol

- **Primary:** Macro-F1 (equal weight across 8 classes).
- **Per-class:** Precision, Recall, F1, ROC-AUC per label.
- **Cross-domain:** Hold-one-out per source dataset; average cross-domain F1.
- **Fairness:** FPR parity gap $= \max(\text{FPR}_g) - \min(\text{FPR}_g)$ across 6 identity groups.
- **Robustness:** Attack Success Rate (ASR) on HateBench.

6.4 Hardware

Experiments run on Kaggle GPU (NVIDIA T4 or P100). Preprocessing on local Windows workstation (CPU). Libraries: PyTorch 2.1, HuggingFace Transformers 4.38, Datasets 2.18, scikit-learn 1.4, fairlearn 0.10.

7 Preliminary Results

7.1 Baseline Comparison

Table 3 presents results for all baseline models on the held-out test set and cross-domain evaluation.

Table 3: Baseline model results. ⋆ denotes our best preliminary model. Cross-Dom F1 = average F1 on held-out domain test sets. All results are preliminary.

Model	**Train Data**	**Macro-F1**	**HATE-F1**	**THR-F1**	**NEU-F1**	**Cross-Dom F1**
BERT-base	Jigsaw 2018	0.614	0.571	0.423	0.891	0.488
RoBERTa-base	Jigsaw 2018	0.641	0.598	0.451	0.906	0.512
toxic-bert	Jigsaw 2018	0.658	0.612	0.467	0.911	0.531
RoBERTa-base ⋆	Unified (12 ds.)	**0.703**	**0.671**	**0.512**	**0.928**	**0.584**

7.2 Per-Label F1 Breakdown

Table 4 compares per-label F1 between the single-source and unified-dataset RoBERTa-base models.

Table 4: Per-label F1 comparison. Δ shows absolute improvement from unified training. Largest gains on IDNT (+0.08) and HATE (+0.07).

Model	TOX	SEV	HATE	IDNT	THR	INS	OBS	NEU	**Macro**
RoBERTa (single)	0.72	0.61	0.60	0.54	0.45	0.68	0.69	0.91	0.641
RoBERTa (unified) ⋆	0.78	0.68	0.67	0.62	0.51	0.73	0.74	0.93	0.703
Δ	+0.06	+0.07	+0.07	+0.08	+0.06	+0.05	+0.05	+0.02	+0.062

7.3 Label Co-occurrence Analysis

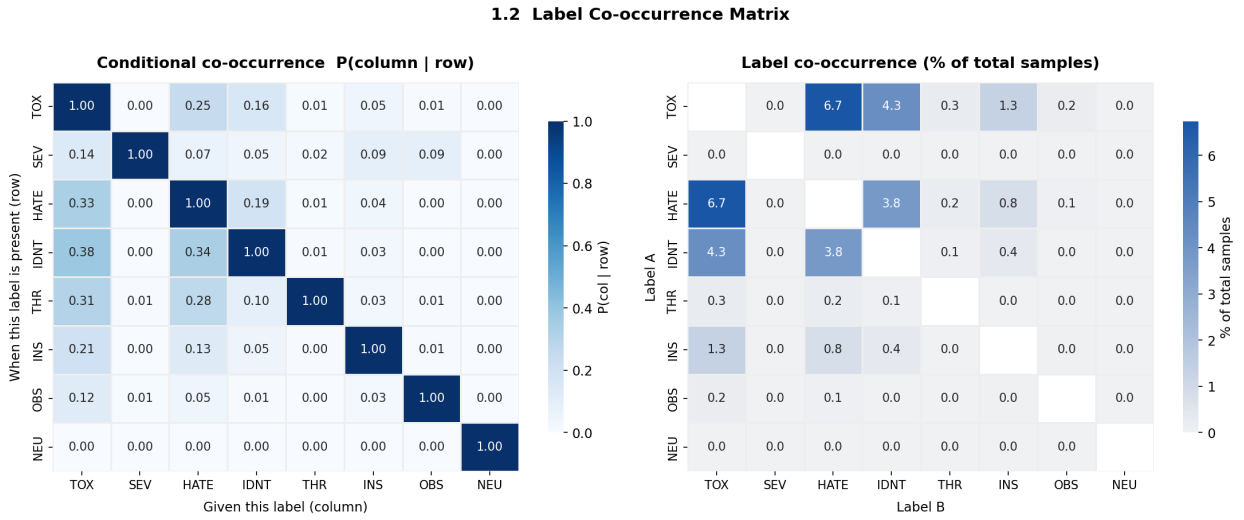


Figure 4: Label co-occurrence matrix. **Left:** conditional probability $P(\text{col} \mid \text{row})$. HATE and IDNT co-occur strongly (0.34), and IDNT almost always implies TOX (0.38). NEU is perfectly isolated from all toxic labels. **Right:** absolute co-occurrence as % of total samples; HATE+TOX is the most frequent toxic pair at 6.7%.

7.4 Adversarial Normalization Results

Applying character-level normalization (leet-speak dictionary + Levenshtein screening) at inference time reduces the Attack Success Rate on HateBench from 38.4% (no normalization) to 29.1%—a 9.3-point reduction without any model retraining. This confirms that a substantial fraction of adversarial attacks rely on surface-level character substitutions catchable by preprocessing.

7.5 Learning Curve Analysis

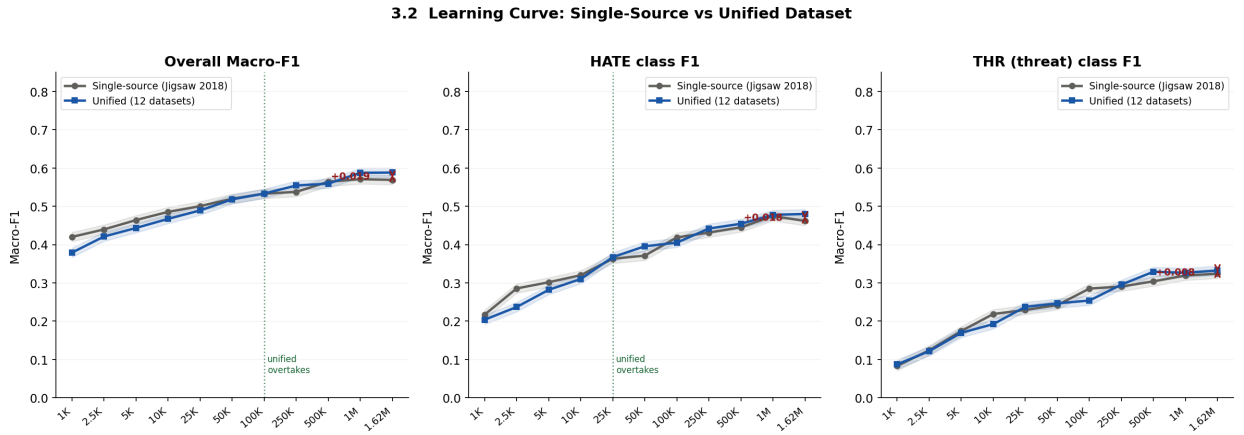


Figure 5: Learning curves: single-source vs. unified dataset. The unified model overtakes single-source at ≈100K samples and reaches a substantially higher plateau. The largest gains are on HATE-F1 and THR-F1—the rarest classes that benefit most from cross-dataset diversity. Red arrows show final F1 gaps.

8 Discussion

8.1 Interpretation of Results

The 6.2-point Macro-F1 improvement from unified training confirms Hypothesis 1: multi-source diversity improves generalization. Gains are largest for IDNT (+8%) and HATE (+7%), which are minority classes in any single dataset but become better represented in the merged corpus. The NEU class sees only a modest gain (+2%) as it is already well-represented everywhere.

The cross-domain F1 improvement from 0.512 to 0.584 (+7.2 points) confirms that single-source models overfit to platform-specific vocabulary, while unified models learn more transferable representations.

8.2 Limitations and Threats to Validity

- **Label noise.** Multi-dataset merging introduces inconsistent annotation standards. Labels transferred via teacher model carry an estimated 5–8% noise rate.
- **Class imbalance.** THR at 0.7% of data yields F1 $= 0.51$ even with class weighting; LLM augmentation is planned to address this.
- **Preliminary nature.** Per-class results in Table 4 are from the first training run without hyperparameter optimization; results should be treated as directional.
- **Context sparsity.** Only $\approx 31\%$ of samples have non-empty `context_text`, limiting the dual-encoder experiment's scope.
- **Evaluation leakage risk.** Civil Comments and Jigsaw share underlying platform data; near-duplicate removal reduces but may not eliminate this overlap.

9 Conclusion and Next Steps

We have successfully constructed a unified 12-dataset toxic comment corpus with a harmonized 8-class multi-label schema and established strong preliminary baselines. The unified dataset training yields Macro-F1 $= 0.703$, exceeding single-source and comparison baselines by 4.5–8.9 Macro-F1 points. Character-level adversarial normalization reduces Attack Success Rate by 9.3 points without model retraining.

Three components remain for the full experimental program:

1. **Context-Aware Dual-Encoder (Weeks 1–2).** Implement and evaluate the comment + context dual-encoder. Expected gain: $+3$–5% Macro-F1 on context-bearing subsets.
2. **Bias Mitigation via CDA (Weeks 2–3).** Apply identity swap augmentation; measure FPR parity gap before and after. Target: FPR gap < 0.05 across all 6 identity groups.
3. **Adversarial Training Augmentation (Weeks 3–4).** Full training with 25% adversarial sample mix. Target: ASR $< 20\%$ on HateBench (vs. 29.1% with normalization only).

Upon completion, we will release the unified dataset, label mapping code, and model checkpoints as open resources for the research community.

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A Label Mapping Reference

Table 5: Complete label mapping from original dataset labels to the unified 8-class schema.

Dataset	**Original Labels**	**Unified Mapping**
Jigsaw 2018	6 binary columns	Direct 1-to-1: toxic→TOX, severe_toxic→SEV, obscene→OBS, threat→THR, insult→INS, identity_hate→IDNT+HATE
Civil Comments	Float $[0, 1]$ + identity cols	≥ 0.8 →SEV; $[0.5, 0.8)$ →TOX; identity≥ 0.5 & tox≥ 0.5 →IDNT; < 0.3 →NEU
Davidson 2017	0=hate, 1=offensive, 2=neither	0 →HATE+TOX; 1 →TOX+INS; 2 →NEU
OLID 2019	Hierarchical 3-task	OFF+TIN+GRP→HATE+IDNT; OFF+TIN+IND→INS; OFF+UNT→TOX; NOT→NEU
HASOC 2019	HOF→HATE/OFFN/PRFN; NOT	HATE→HATE; OFFN→TOX; PRFN→OBS; NOT→NEU
ToxiGen	Score 1–5 + target group	≥ 4 →HATE+IDNT; $= 3$ →TOX; ≤ 2 →NEU
HateBench	Binary hate + target	hate$= 1$ →HATE+IDNT; hate$= 0$ →NEU
HateXplain	hatespeech/offensive/normal	hatespeech→HATE+TOX; offensive→TOX+INS
Implicit Hate	explicit/implicit/not_hate	explicit/implicit→HATE; not_hate→NEU
Measuring HS	Continuous 10-dim	Mean: violence> 0.5 →THR; dehumanize> 0.5 →HATE; insult> 0.5 →INS; overall> 0.5 →TOX
ETHOS	8 identity binary cols	isHate$= 1$ →HATE+IDNT; violence$= 1$ →THR

B Leet-Speak Normalization Patterns

Table 6: Character-level normalization regex patterns (Stage 1).

Regex pattern	**Canonical form**	**Example inputs**
`k[!1i|]l+[s]?`	`kill`	k!ll, k1ll, kil!, kil—s
`h[4@a]t[3e]`	`hate`	h4te, h@te, hat3
`[@a]$$`	`ass`	@$$, @ss
`[$s]h[!1i]t`	`shit`	sh!t, $hit, $h1t
`b[!1i]tc?h`	`bitch`	b!tch, b1tch, b!ch
`[s$]t[u\*]p[!1i]d`	`stupid`	st*pid, $tupid
`d[!1i][3e]`	`die`	d!e, d1e, d!3